

Lecture Material #8

Information Security

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Advanced Encryption Standard

AES





AES Algorithm in Cryptography

Plain Text transform in Matrix Form

For Example, “AES USES A MATRIX”

Text	A	E	S	U	S	E	S	A	M	A	T	R	I	X	Z	Z
------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Hexadecimal	00	04	12	14	12	04	12	00	0C	00	13	11	08	17	19	19
-------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

$$\begin{bmatrix} 00 & 12 & 0C & 08 \\ 04 & 04 & 00 & 17 \\ 12 & 12 & 13 & 19 \\ 14 & 00 & 11 & 19 \end{bmatrix} \text{ State}$$





AES Algorithm in Cryptography

	DEC	HEX		DEC	HEX
A	00	00	N	13	0D
B	01	01	O	14	0E
C	02	02	P	15	0F
D	03	03	Q	16	10
E	04	04	R	17	11
F	05	05	S	18	12
G	06	06	T	19	13
H	07	07	U	20	14
I	08	08	V	21	15
J	09	09	W	22	16
K	10	0A	X	23	17
L	11	0B	Y	24	18
M	12	0C	Z	25	19

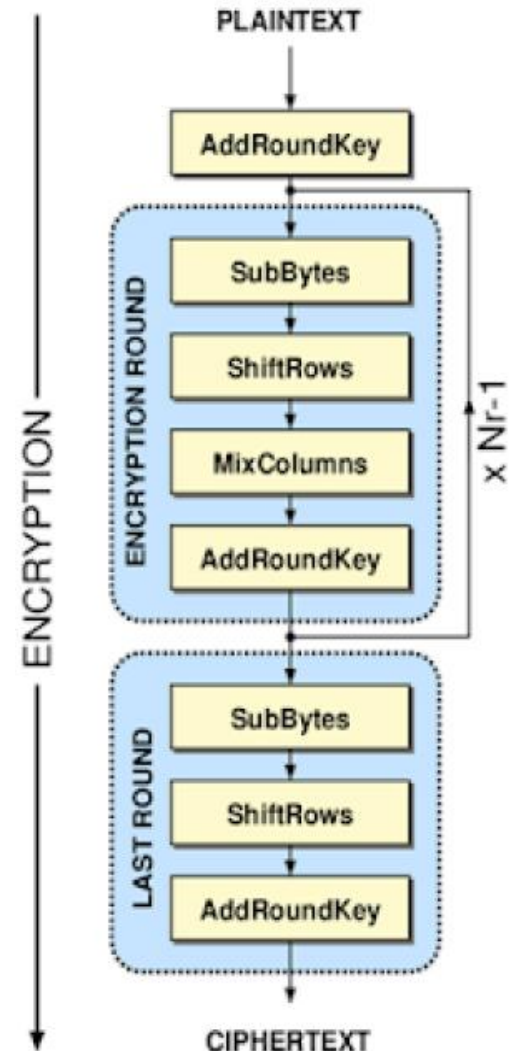




AES Algorithm in Cryptography

The number of **rounds** is **10**, for the case when the encryption key is 128 bit long. (12 rounds – 192 bits, 14-rounds – 256 bits).

For encryption, each round consists of the following four steps: **SubBytes**, **ShiftRows**, **MixColumns**, **AddRoundKey**





AES Algorithm in Cryptography

SubBytes / Substitute Bytes

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

For example, EA → 87

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5



87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6





AES Algorithm in Cryptography

Shift Row Transformation

The shift row transformation is called ShiftRows.

Rules of shifting rows,

Row 1 $\Rightarrow$ No Shifting

Row 2 $\Rightarrow$ 1 byte left shift

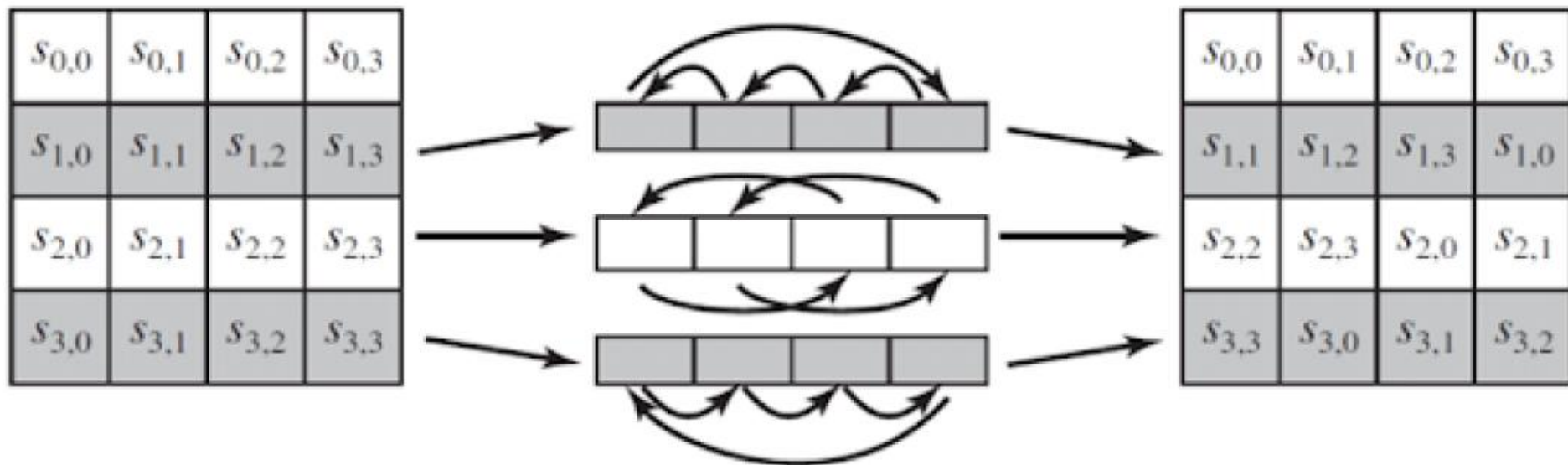
Row 3 $\Rightarrow$ 2 byte left shift

Row 4 $\Rightarrow$ 3 byte left shift





AES Algorithm in Cryptography





AES Algorithm in Cryptography

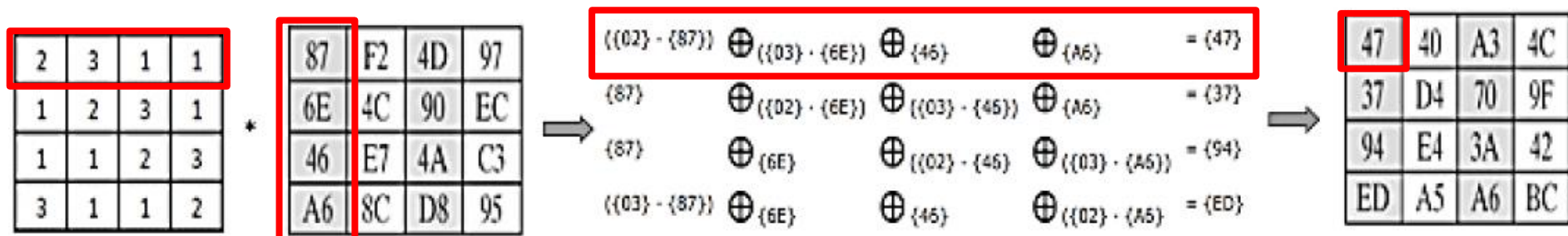
Mix Columns

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix} \Rightarrow \begin{aligned} s'_{0,j} &= (2 \cdot s_{0,j}) \oplus (3 \cdot s_{1,j}) \oplus s_{2,j} \oplus s_{3,j} \\ s'_{1,j} &= s_{0,j} \oplus (2 \cdot s_{1,j}) \oplus (3 \cdot s_{2,j}) \oplus s_{3,j} \\ s'_{2,j} &= s_{0,j} \oplus s_{1,j} \oplus (2 \cdot s_{2,j}) \oplus (3 \cdot s_{3,j}) \\ s'_{3,j} &= (3 \cdot s_{0,j}) \oplus s_{1,j} \oplus s_{2,j} \oplus (2 \cdot s_{3,j}) \end{aligned}$$

Predefine Matrix

State Array

New State Array



$$(02 \cdot 87) \oplus (03 \cdot 6E) \oplus (01 \cdot 46) \oplus (01 \cdot A6)$$





AES Algorithm in Cryptography

Mix Columns

$$(02 \cdot 87) \oplus (03 \cdot 6E) \oplus (01 \cdot 46) \oplus (01 \cdot A6)$$

Multiplier	Rule
01 × x	x
02 × x	Left shift; if MSB=1 → XOR 1B
03 × x	(02 × x) ⊕ x





AES Algorithm in Cryptography

Mix Columns

$$(02 \cdot 87) \oplus (03 \cdot 6E) \oplus (01 \cdot 46) \oplus (01 \cdot A6)$$

$$04 \times x = 02 \times (02 \times x)$$

$$08 \times x = 02 \times (04 \times x)$$

$$10 \times x = 02 \times (08 \times x)$$

AES bytes represent **polynomials of degree ≤ 7** .

All calculations must stay within this range.





AES Algorithm in Cryptography

Mix Columns

$$(02 \cdot 87) \oplus (03 \cdot 6E) \oplus (01 \cdot 46) \oplus (01 \cdot A6)$$

$$87 = 10000111$$

Left shift $\rightarrow$ 00001110 (0E) with overflow

$$0E \oplus 1B = 15$$

$$01 \times 46 = 46$$

$$01 \times A6 = A6$$

$$02 \times 6E$$

$$6E \ll 1 = DC$$

$$03 \times 6E = DC \oplus 6E = B2$$

$$15 \oplus B2 \oplus 46 \oplus A6$$

$$15 \oplus B2 = A7$$

$$A7 \oplus 46 = E1$$

$$E1 \oplus A6 = 47$$





AES Algorithm in Cryptography

47 = 0100 0111

AC = 1010 1100

----- XOR

EB = 1110 1011

AddRoundKey

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

$\oplus$

AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

=

EB	59	8B	1B
40	2E	A1	C3
F2	38	13	42
1E	84	E7	D2



